

LLM-Modulo Semantic-Numeric CEGIS

Current architecture, evidence boundaries, and trajectory-generation scope

Updated 2026-08-21

Revision scope. The first version of this module already had a typed hypothesis population, one learner per hypothesis, a trajectory membership Oracle, and an outer fit-query-evidence-revise loop. The architecture was not rebuilt from zero. It was progressively hardened after failures exposed weak LLM output, overlapping hypotheses, heuristic falsifiers, excessive Oracle pressure, label leakage, unstable numeric fitting, and proxy structures that could win for the wrong reason.

Current invariant: the LLM proposes semantic structure and experiments. Demonstrations and whole-trajectory Oracle labels determine the numeric boundary. The LLM never receives an obstacle center, radius, state-level collision mask, or private evaluation metric.

1 Problem and information contract

The system receives:

- a task description d and a public variable schema $\mathcal{S}$;
- safe expert trajectories $\mathcal{D}_E = \{\tau_i^E\}$;
- a capability-limited trajectory membership Oracle $\mathcal{O}$.

A trajectory is $\tau = (x_1, \dots, x_T)$. The Oracle returns one label for the complete trajectory,

$$\mathcal{O}(\tau) \in \{0, 1\}, \quad 0 = \text{SAFE}, \quad 1 = \text{VIOLATION}. \quad (1)$$

It does not identify a violating state or expose hidden geometry.

The semantic reasoner initially receives only task text and the public schema. It does not receive raw demonstrations. Demonstrations enter the numeric learner, warmup generator, and falsifier. In later rounds, the reasoner receives a compact public evidence report, not raw private evaluation.

The benchmark controller still holds a private evaluator. It may compute IoU and related diagnostics after the semantic decision of each round and after optional finalization. These values are written to evaluation-only artifacts and never enter fitting, acquisition, evidence, prompts, revision, or champion scoring. This is code-path non-use, not process-level physical isolation.

2 Current control flow

In the original Obstacle2D path, global acquisition means global selection from one shared cross-hypothesis candidate pool; its candidates remain expert-anchored. The later CarryWaterActive adapter adds a separate pool of independently generated public-dynamics rollouts, described below.

In each outer round the controller:

1. fits each active structure on the complete trainable buffer;
2. freezes validation and pre-query predictions;
3. generates a candidate pool from a query-priority beam and pending interventions;
4. spends one Oracle budget sequentially, without retraining inside the batch;

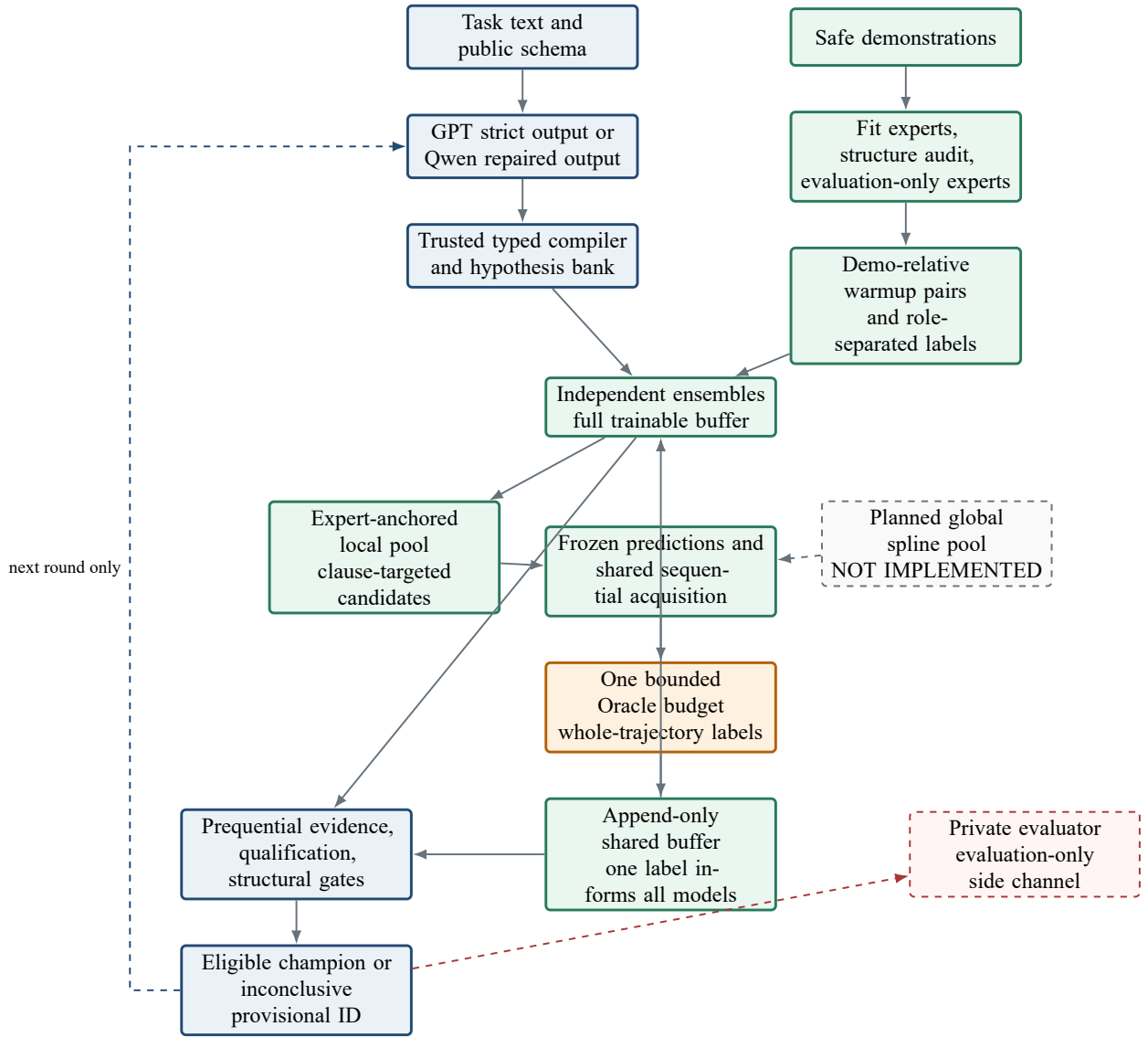


Figure 1: Obstacle2D loop. The dashed gray direct-spline generator is a proposed branch for that path, not an implemented capability. The red evaluation path never feeds back into learning.

5. refits, compiles evidence, applies qualification and structural gates;
6. returns an eligible champion, or an inconclusive provisional ID;
7. calls semantic revision only if another numeric round remains;
8. optionally finalizes a qualified champion using disjoint calibration and selection roles.

3 Typed hypotheses and multiple constraints

An atomic clause is a typed object:

```

Clause = {
  variables,      # public observable features
  coupling,      # joint | independent
  relation,      # forbidden_region | upper_bound
                # | lower_bound | equality_band
  temporal_operator, # max | mean | last
  model_family, clause_id # mlp | linear; stable identity
}

```

The trusted compiler turns every field into a different computation graph. Joint coupling sends all selected variables to one head. Independent coupling builds one head per variable and joins violation scores with max. Scalar upper/lower bounds use learnable thresholds. Equality bands and forbidden regions use their compiled heads. Unknown variables, unsupported combinations, duplicate structures, and malformed actions are rejected.

A composite hypothesis currently uses `any_violation` semantics:

$$G_H(\tau) = \max_{c \in H} G_c(\tau). \quad (2)$$

Thus a trajectory is safe only when every clause is safe. A feature-A equality and a feature-B inequality have separate clause IDs and heads. For a composite false-unsafe probe, generation, hard certification, refinement, query gating, deduplication, and cross-round scheduling all preserve the target clause identity. A nuisance clause cannot receive credit for the requested clause’s crossing.

GPT uses strict Structured Outputs. Qwen uses conservative surface repair followed by the same compiler. Successful responses and parse outcomes are auditable. Backend failures default to fail-fast unless an explicit fallback experiment is enabled. The small Qwen backend is still not claimed to provide reliable multi-round reasoning.

4 Data roles and numeric fitting

4.1 Leakage-controlled roles

Known-safe demonstrations are separated into fit experts, structure-audit experts, and held-out evaluation experts. Warmup queries are grouped by complete pair families and assigned to:

- `gradient training`;
- `warmup_validation`, used as prequential selection evidence;
- `final_calibration`, used only for optional threshold calibration.

No warmup pair family is split across roles. Calibration and selection records do not enter gradient fitting. Structure-audit experts are a separate known-safe set; they are not warmup validation records.

4.2 Full-buffer ensembles

The early learner used per-member bootstrap samples and could omit a large fraction of a small paid query set. The selected default is now `bootstrap_queries=false`: every ensemble member sees every trainable Oracle record, while independent initialization preserves ensemble diversity.

For a source-matched Oracle violation derived from a uniquely resolved safe anchor, max-MIL excludes selected-feature states whose representation is unchanged from the anchor. This blocks the repeated safe start or endpoint from becoming a cheap violation witness. The mask is not a ground-truth collision-time label. It is source-specific, and unresolved or representation-invariant records fall back to the original bag loss. Mean and last temporal operators retain their own aggregation semantics.

Training uses a differentiable smooth surrogate, while inference, pre-query evidence, and hard certificates use the compiled hard temporal score. A trajectory label remains a multiple-instance label: a violation bag contains at least one violating state, but the learner is not told which one.

5 How Obstacle2D candidate trajectories are generated

5.1 Obstacle2D candidates are expert-anchored

Let a selected expert path be $E = (e_0, \dots, e_{T-1})$ and let its endpoint chord be

$$L_t = (1 - s_t)e_0 + s_te_{T-1}. \quad (3)$$

Warmup creates deterministic demonstration-relative pairs:

$$X_{\text{toward}} = E - \alpha(E - L), \quad X_{\text{continue}} = E + \alpha(E - L). \quad (4)$$

The outer falsifier initializes a path as

$$X^{(0)} = (1 - \mu)E + \mu L + a \sin(\pi s) \sin(k\pi s)d, \quad (5)$$

where restarts vary direction d , frequency k , and interpolation μ . The optimizer freezes the constraint model and changes only interior waypoints. Start and goal are restored exactly. All modes include expert-distance, smoothness, workspace, and step penalties.

Table 1: Implemented candidate modes and their guarantees

Mode	Current meaning and validation
shortcut	Strongly interpolate the expert detour toward its endpoint chord and optimize toward model-safe. Heuristic plus kinematic validation.
model_false_safe	Search for a path the source model accepts but the Oracle may reject. Heuristic plus kinematic validation.
boundary_uncertainty	Search near a model boundary and model disagreement. It has no false-unsafe hard certificate.
local_feature_stress	Stress one selected feature while penalizing changes to preserved features. Heuristic plus kinematic validation.
model_false_unsafe	Start from a source-model-safe expert, cross the full model and target clause with calibrated margins, then refine to a nearby frozen-model boundary.

5.2 False-unsafe hard certification

Only false-unsafe receives the complete hard-certificate chain. With decision threshold θ :

- the expert anchor must satisfy source score $\leq \theta - 0.02$;
- generation must satisfy both full-model and target-clause score $\geq \theta + 0.05$;
- every waypoint obeys $\max_t \|X_t - E_t\|_2 \leq 0.32$ under the selected default;
- a kinematically valid crossing checkpoint is retained even if later optimization steps regress;
- a 32-point homotopy scan plus boundary refinement finds the first safe-to-unsafe bracket;
- the actual query point retains a separate margin of 0.02 and names the source causal rejector.

If the source rejects the known-safe anchor, the requested clause is uncontrollable, or the certificate is missed, the result remains diagnostic and cannot consume Oracle budget. A certified probe is still only a model crossing. It is not proof that the Oracle will label the trajectory safe, nor proof of physical feasibility.

The five-seed comparison selected one pointwise radius 0.32 over the ladder 0.04/0.08/0.16/0.32. Both arms produced zero Oracle-safe hits among 15 queried false-unsafe probes; median queried deformation did not improve, while optimizer launches increased from 130 to 424. The choice is computational, not an accuracy claim.

6 Shared acquisition and Oracle use

The first outer round normally lets all active structures contribute candidates. Later rounds use a query-priority beam plus any pending semantic intervention targets. Each source may generate several modes, but none receives an automatic Oracle call.

All active models first produce frozen predictions, scores, and uncertainties for every candidate. The acquisition score combines cross-hypothesis disagreement, boundary proximity, ensemble uncertainty, novelty, and source potential. Duplicate or non-queryable candidates are removed.

The Oracle budget is then spent sequentially. After each answer, the controller may update only trainable safe/violation deficits, empirical intervention yield, and confirmed causal-safe signatures. Models are not retrained until the batch ends. Safe and violation balance-seeking together share a cap of two slots; a safe-seeking slot additionally requires the false-unsafe causal certificate. One paid label is stored with every active model's frozen prediction, so it can support or refute the entire bank.

This design reduces the old tendency toward one-query-per-hypothesis. It does not eliminate the membership Oracle assumption, and it does not prove that 26 calls are universally minimal.

7 Evidence, qualification, structural gates, and finalization

The Evidence Compiler uses frozen pre-query predictions, first-use warmup-validation predictions, structure-audit expert safety, and the append-only query log. It does not rescore training labels with a model that has already fitted them.

Selection is two-stage:

1. **Qualification gates** enforce minimum sample counts, safe accuracy, violation recall, fit-expert safety, and structure-audit safety.
2. **Ranking** compares only eligible candidates with a configuration-weighted engineering score. If none is eligible, the run reports `selection_status=inconclusive` and keeps only a provisional ID for continued querying.

Structural checks are deliberately separated by logical strength:

- **Exact contradiction:** a pure-last representation carrying both safe and violation labels is ineligible.
- **Exact contradiction:** an atomic 1D/2D joint-linear-max `forbidden_region` is ineligible when Oracle-violation points lie in the convex support of at least two distinct safe anchors. This gate changes eligibility, not acquisition scores.
- **Evidence priors, not theorems:** nested-feature, progress-proxy, and dynamics-proxy structures must show a configured material gain over an appropriate simpler competitor.

General ranking among several structurally viable eligible families is not solved. The terminal-round selector still uses the raw configured score among eligible models; adaptive intervention yield, implementation capacity, and uncalibrated cross-family uncertainty remain imperfect comparison terms.

Optional finalization freezes the selected structure, fits candidate numeric states without mutating the incumbent, calibrates thresholds on `final_calibration`, selects once on disjoint `warmup_validation`, and commits only a candidate that still passes public gates. Ordinary GPT/Qwen profiles do not all enable finalization; the validated diagnostic profile does.

8 Why the old high-y proxy could win

Temporal max aggregates a violation score, not the raw feature peak. For a lower-bound clause with threshold c and positive scale s ,

$$g_t = s(c - y_t), \quad G(\tau) = \max_t g_t = s(c - \min_t y_t). \quad (6)$$

A genuinely strict “always high y ” rule is therefore contradicted by a sufficiently low start or goal. The failed y -only model did not mathematically ignore the endpoint. It could instead move c low enough, or shrink its score, to become nearly all-safe. Sparse whole-trajectory evidence and the early selector did not impose a strong enough violation-recall gate, so a simple but vacuous proxy could win.

The current system directly blocks the all-safe pathway with class-specific qualification, role-separated warmup evidence, complete trainable-buffer fitting, and explicit inconclusive status. Pure-last and one affine-max family also receive exact representation tests. These changes prevented recurrence in the tested runs, but they are not a universal causal-identification theorem.

With finite whole-trajectory labels, a y -only proxy can still survive if every queried trajectory agrees with it. A more complete Obstacle2D treatment needs representation-matched counterfactuals: same endpoint and the same y sufficient statistic, but different joint spatial path. Opposite Oracle labels would then provide an exact collision for the y -only representation. This Obstacle2D generator is not implemented yet; CarryWaterActive supplies analogous exact collisions for its own proxy families.

9 What the experiments establish

Table 2: Evidence from distinct experiment stages; rows must not be merged into one run

Stage	Setting	Supported conclusion
Live GPT integrated	One live structured-output run; 26 Oracle calls, 3 GPT interactions, zero fallback	Correct spatial champion, IoU 0.8146, held-out expert-safe 1.0. Demonstrates an executable live semantic path, not multi-seed stability.
Single vs ladder	Frozen GPT round-0 bank, zero LLM, five paired numeric seeds	The ladder did not improve safe yield or median deformation and used about $3.26\times$ optimizer launches; use single 0.32 by default.
Numeric fitting	Frozen bank, classic bootstrap vs full all-state vs full source-anchor mask	Spatial champion counts were 2/5, 3/5, and 4/5; under the mask spatial itself was eligible in 5/5. Shows fitting stability, not live semantic quality.
Linear-max gate	New five seeds, audit-only vs enforced, otherwise paired	Both arms already produced qualified spatial champions in 5/5. The gate is adopted as a sound no-regression guard; this experiment does not show a champion-rate or private-IoU gain.

Stage	Setting	Supported conclusion
Regression suite	Current source tree	115/115 unit tests pass, including composite clauses, hard margins, acquisition state, role separation, source-anchor masking, exact gates, 12D dynamics replay, privacy contracts, and proxy collisions.

Obstacle2D remains primarily a mechanism and diagnosis benchmark. It shows that the modules execute and exposes proxy failures. A fixed (x, y) MLP or hand-designed circular family can solve much of the same boundary task, so this benchmark alone cannot establish the full advantage of LLM-guided semantic revision.

10 Obstacle2D trajectory-space limitation

Every Oracle-query candidate in the original Obstacle2D path is expert-anchored. This is useful locally: it supplies a plausible path and a known-safe reference for false-unsafe. It also biases support toward demonstrated detours and their endpoint chords. Smooth basis restarts may move into another region, but they do not guarantee a new upper/lower route or a new homotopy class.

The proposed `global_direct_spline` branch is **not implemented**. A valid future design would use only an explicit public TaskSpec containing start, goal, horizon, workspace, and public kinematic limits. It would sample low-dimensional control points, interpolate a smooth spline, filter workspace/step/curvature violations, and rank unlabeled proposals using frozen-model disagreement, boundary proximity, novelty, and structure discrimination. If start, goal, or horizon still come from a demonstration, the accurate claim is “does not use expert interior waypoints,” not fully expert-independent generation.

The generator must not read hidden obstacle geometry and must not invent labels. Global candidates would compete with local candidates under the same Oracle budget. Because they have no known-safe anchor, they must not automatically consume safe-balance slots.

The next controlled experiment should compare:

1. **local-only**: the current expert-anchored generator;
2. **global-only**: public TaskSpec plus direct spline candidates;
3. **hybrid**: both pools under one shared acquisition budget.

Bank, numeric seeds, role splits, candidate-compute budget, and Oracle calls must be paired. Primary outcomes should be structure recovery, proxy rejection, new path-family coverage, query duplication, and information per Oracle call. Private IoU remains a post-selection diagnostic. If hybrid does not produce stable public evidence gains, the simpler local-only system should remain the default.

11 CarryWaterActive: independent rollouts and heterogeneous constraints

CarryWaterActive is a task-specific 12D benchmark added after the preceding Obstacle2D analysis. It does not perturb expert observations. A public generator samples new task contexts and control profiles, derives $[T - 1, 6]$ actions, and replays every trajectory through one public dynamics implementation. The learner receives 40 fit experts, 12 validation experts, 12 structure-audit experts, public dynamics, and a pool of 512 unlabeled independent rollouts.

The intended hidden structure is a conjunction of three typed atomic clauses under any-violation composition: an always-on equality band for relative requested height `target_dz`, an upper bound on direction-independent 3D speed, and an upper bound on total tilt from vertical. Yaw is intentionally free. This exercises the mixed equality-plus-inequality case without collapsing it into one ambiguous all-feature relation.

All hypotheses score the same unlabeled pool before one shared acquisition policy spends the membership-query budget. Multiple models can propose or score candidates, but they do not receive one Oracle call each. Warmup members are removed from later acquisition, and the planar 0.32 trust-radius setting is not reinterpreted as a physical radius in the 12D task.

Private evaluation is resolved only after the training result and checkpoint are frozen. It contains 512 matched safe/violation pairs, or 1,024 trajectories, balanced across height-only, speed-only, tilt-only, and multi-clause failures. Three exact representation collisions make common shortcuts falsifiable:

- height pairs share the complete world- z sequence but have opposite labels because the requested height differs;
- speed pairs share one complete velocity component but differ in 3D speed through another axis;
- tilt pairs share the complete yaw sequence but differ in roll/pitch tilt.

Evaluation therefore reports worst pair-target balanced accuracy, exact-pair accuracy, pair-ranking accuracy, minimum clause recall, and per-clause recall in addition to aggregate trajectory metrics.

The deterministic 13-query connectivity smoke completed with 3 safe and 10 violation labels and did not load private evaluation before freeze. It returned `selection_status=inconclusive`: the provisional world- z proxy was not champion-eligible. Post-hoc private balanced accuracy was 0.500, worst pair-target balanced accuracy was 0.500, and exact-pair accuracy was 0.000. Diagnostic evaluation of all seven frozen-bank learners also showed that the correct composite was not numerically fitted under this tiny budget. The result proves pipeline connectivity and proxy-failure visibility, not benchmark superiority.

An authorized GPT semantic-only smoke accepted four of four schema-valid hypotheses with zero fallback, zero augmentation, and zero Oracle queries. Its first hypothesis contained exactly the relative-height equality band, 3D-speed upper bound, and total-tilt upper bound while leaving yaw free. This isolates the remaining failure to numeric evidence and selection for this run; it does not establish cross-seed GPT stability.

The next minimal experiment freezes the public instance and semantic bank and runs five pre-registered numeric seeds at a total Oracle budget of $Q = 48$, including warmup. If the performance-query curve is still unsaturated, $Q = 64$ must be registered before inspecting that extension. Primary outcomes are qualified exact structure recovery, abstention, exact-pair accuracy, worst-clause recall, safe counterfactual yield, and cost per Oracle call.

12 Assumptions, remaining limits, and code map

The method still assumes three user-facing information channels: task description, safe demonstrations, and near-binary judgments of selected new trajectories. There is no abstain or noise model. False-unsafe validity checks endpoints, workspace, finite values, and maximum step; it does not prove robot dynamics or task feasibility. Positive multi-seed learning evidence still comes from one 2D benchmark and one frozen GPT bank. CarryWaterActive adds a validated 12D benchmark and one honest failure smoke, not yet a cross-task significance result.

Table 3: Main implementation modules

Module	Main job
<code>hypotheses.py</code>	Typed atomic/composite IR, compiler, bank, clause validation, and revision actions.
<code>semantic.py</code>	GPT/Qwen backends, trusted parsing/repair, frozen-bank replay, and deterministic evidence policy.
<code>data.py</code>	Public feature schema, differentiable derived features, normalization, and bounds.
<code>learner.py</code>	Independent ensembles, temporal aggregation, full-buffer fitting, and source-anchor MIL masking.
<code>falsifier.py</code>	Demo-relative warmup, expert-anchored optimization, hard checkpoints, and kinematic validation.
<code>pool_falsifier.py</code>	Shared selection from independently generated, unlabeled task-specific rollout pools.
<code>carrywater_active.py</code>	Public 12D dynamics replay, inverse-action consistency, and trajectory validation.
<code>loop.py</code>	Query beam, candidate pool, boundary refinement, sequential acquisition, evidence flow, revision, finalization, and artifacts.
<code>evidence.py</code>	Prequential evidence, qualification, terminal/linear exact gates, and heuristic minimality gates.
<code>oracle.py</code>	Capability-limited whole-trajectory membership interface.
<code>evaluation.py</code>	Private metrics and plots on the evaluation-only side channel.

Artifacts include the resolved configuration, source/input/runtime fingerprints, semantic interactions, hypothesis bank, query diagnostics, Oracle query archive, evidence and evaluation histories, stage/finalization diagnostics, all final model states, and plots. They do not preserve every unqueried candidate trajectory or every intermediate checkpoint, so a single run supports decision auditing but not perfect replay of all internal optimization.

Bottom line: the architecture now includes a heterogeneous 12D benchmark and a non-expert-perturbation rollout pool. The first tiny-budget run correctly abstains but does not recover the composite constraint. Publication claims require the pre-registered multi-seed, equal-budget comparison; finite trajectory labels still do not guarantee causal recovery.